ACCELERATOR DESIGN EDUCATIONAL PRIMER – CONCEPTUALIZING AND OPTIMIZING THE HYBRID LHEC-LIKE ELECTRON-ION COLLIDER DESIGN*

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Abstract

The Electron-Ion Collider (EIC) Mission Need requires $\sqrt{s}$ = 20–100 GeV (upgradable to 140 GeV) and luminosity 10^{33}–10^{34} cm^{-2} s^{-1}. The current ring–ring baseline achieves the full scope, including ~10^{34} cm^{-2} s^{-1} across all energies. However, when the design is re-optimized for the lower boundary – accepting 10^{33} cm^{-2} s^{-1} and prioritizing cost – an alternative configuration emerges as more advantageous: a hybrid LHeC-like electron accelerator using multi-pass energy recovery linacs (ERL). This solution reduces electron-beam power by roughly an order of magnitude, yielding nearly a factor of two reduction in total project cost compared with the present baseline while still satisfying the minimum physics requirements. The study performs parametric cost and performance modeling, augmented by AI-driven optimization, to explore this design space. Serving primarily as an educational exercise for the next generation of accelerator physicists and engineers, the paper demonstrates modern design methods: rapid parametric scans, cost-driven optimization, and integration of AI tools. It examines technical feasibility, identifies critical R&D (high-current ERL operation, beam–beam effects, synchronization, etc.), and discusses how such re-optimization studies can be used to train designers in an era when artificial intelligence dramatically expands exploration of complex accelerator parameter spaces.

PREAMBLE – AI AND EDUCATION

The emergence of AI-assisted coding presents both unique challenges and unprecedented opportunities for accelerator physics education. Institutions like Fermilab (USPAS) and CERN (CAS) need to rapidly adapt their pedagogical approaches, and the Genesis program [1] in the US further encourages AI for accelerated accelerator design. A natural vehicle for developing design skills is the focused mini-project such as Refs. [2, 3], where students tackle a realistic challenge in a short intensive period. This paper rehearses what such a mini-project looks like when AI tools are available at every stage – from parameter optimization and code generation to cost estimation – serving as a draft script for future AI-assisted USPAS courses.

* Parameters of the presented concepts were developed with assistance from Grok AI, Claude AI and Gemini AI.

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EIC-LHEC MINI-PROJECT

The mini-project assignment is: *Given the EIC proton beam stored in the RHIC-derived Hadron Storage Ring, design the simplest and most cost-effective electron accelerator complex satisfying the EIC CD0 Mission Need lower boundary –* $\sqrt{s} \approx$ *29–100 GeV and* $\mathcal{L} \sim 10^{33}$ *cm*$^{-2}$*s*$^{-1}$ *– and estimate its cost relative to the ring-ring baseline.*

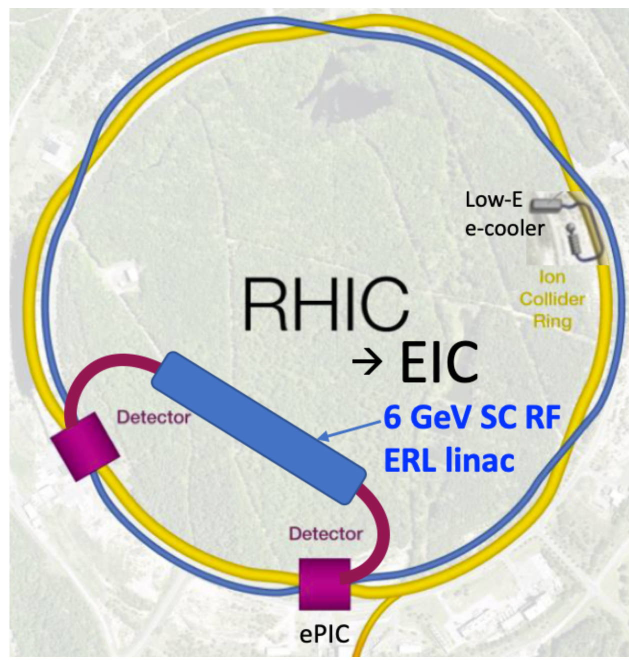

Figure 1: Schematic layout of the EIC built as an LHeC-like hybrid with energy recovery.

The motivation comes from the ESPP Open Symposium [4], where it was argued that inserting a new electron ring into the LHC tunnel for LHeC would be cost-prohibitive and luminosity-limiting [5]. This prompted the question: could replacing the EIC electron storage ring with a superconducting energy-recovery linac (ERL) yield a similar cost-performance advantage? The proposed hybrid keeps the full EIC proton infrastructure intact and replaces the electron ring with a multi-pass ERL: electrons are accelerated in a superconducting radio-frequency (SRF) linac, recirculated through arcs, brought to collision, and then decelerated back through the linac for energy recovery. Key advantages over the ring-ring baseline are: electron beam power reduced by ~10× [each bunch used once, such that a large interaction-point (IP) disruption is acceptable]; no large electron storage ring, substantially reducing civil and magnet costs; and a 6 GeV linac traversed three times covering the full 18 GeV EIC energy range.

The layout is shown in Fig. 1: the ERL linac is installed in a new tunnel in the South-West portion[1] of the RHIC inner area, with recirculation arcs routed through new arc tunnels, delivering the electron beam to the ePIC interaction region.

The mini-project proceeds through the following steps, each carried out with AI assistance (primarily Grok and Claude) and described in subsequent sections: beam parameter and luminosity modeling; beam–beam evaluation; IR optics and magnet technology assessment; energy recovery and recirculation analysis; cost estimation; and R&D identification. At each step, AI tools wrote Python scripts and draft text, while the "students" (here, the authors) formulated physically meaningful prompts and verified outputs against accelerator physics principles.

EIC-LHEC PARAMETERS

Approach and AI prompting strategy. The parameter development followed an iterative dialogue with AI tools, beginning from the two parent designs and converging on a self-consistent hybrid. The initial prompts requested Python codes to compute beam parameters, beam sizes at the IP, luminosity, and beam–beam tune shifts $\xi_{x,y}$ and RMS divergences for the standard EIC and LHeC designs separately, using published parameter tables [6, 7] as inputs. Details of prompts, outputs, and generated codes are available at [8].

Baseline EIC and LHeC reference parameters. Table 1 shows abridged parameter sets for the two parent designs; full tables are in [6, 7].

Table 1: Abridged LHeC Reference Parameters (Phase-1, 20 GeV e on 7 TeV p) and Baseline EIC ($1{\times}10^{34}$ Peak Luminosity, 10 GeV e on 275 GeV p) Parameters

Parameter	**EIC**	**LHeC**
e / p energy, GeV	10 / 275	20 / 7000
$\sqrt{s}$, GeV	105	749
Number of bunches	1160	2808
Collision frequency, Hz	9.1×10^7	4.0×10^7
e bunch population	1.72×10^{11}	1.19×10^{10}
p bunch population	6.9×10^{10}	2.2×10^{11}
e norm. emitt. h/v, µm	391 / 26	22 / 22
p norm. emitt. h/v µm	3.3 / 0.3	2.5 / 2.5
e $\beta^*_{x/y}$ at IP, m	0.45 / 0.06	0.2 / 0.2
p $\beta^*_{x/y}$ at IP, m	0.80 / 0.07	0.35 / 0.35
e $\sigma_{x/y}$ at IP, µm	94.8 / 8.6	10.6 / 10.6
p $\sigma_{x/y}$ at IP, µm	94.9 / 8.6	10.7 / 10.7
e / p beam current, mA	1555 / 1000	60.1 / 1110
p beam-beam $\xi_{x/y}$	0.012 / 0.012	0.001 / 0.001
Peak lumi. $\mathcal{L}$, cm^{-2}s^{-1}	1.05×10^{34}	5.78×10^{33}

Resolving parameter inconsistencies in the hybrid. When asked to combine EIC proton parameters with LHeC electron parameters, the AI correctly identified three fundamental conflicts before producing any code. (i) *Bunch rate mismatch:* the EIC ring (3834 m circumference, 1160 bunches) gives a collision frequency of ~91 MHz, while the LHeC electron linac assumes ~40 MHz; naively combining the two implies ~1.45 A electron current, far beyond the 60 mA LHeC design value. (ii) *Beam size mismatch:* LHeC electron optics ($\beta^*_{x,y} = 0.20$ m, $\varepsilon_n = 22$ µm) were optimized for the ~10 µm LHC proton beam, whereas EIC protons at 275 GeV give $\sigma_x \approx 173$ µm at the same β^*. (iii) *Current limits:* maintaining $N_e \approx 1.19 \times 10^{10}$ at the EIC rate requires ~1.45 A; enforcing 60 mA instead would reduce N_e to $\sim 4 \times 10^9$. The resolution was to reduce the number of proton bunches by a factor of 3 (from 1160 to 386), proportionally increasing bunch population at constant 1.0 A proton current. This lowers the collision frequency to ~30 MHz, restoring a consistent ~58 mA electron current, and the IP optics were re-optimized to match beam sizes ($\beta^*_x = 1.20$ m, $\beta^*_y = 0.05$ m for electrons; $\beta^*_x = 0.12$ m, $\beta^*_y = 0.06$ m for protons). The AI also noted that ξ_e is unconstrained in a linac-ring scheme since electrons are discarded after each pass; only the proton beam–beam parameter was constrained to $\xi_p < 0.01$. A further step explored $\mathcal{L} = 10^{34}$ cm^{-2}s^{-1} via: (i) proton emittance reduction through coherent electron cooling; and (ii) vertical electron emittance reduction by a factor of ten using a magnetized gun and Derbenev's flat-beam transformer [11].

Hybrid EIC-LHeC parameter sets. Two configurations are given in Tables 2 and 3, both for $\sqrt{s} \approx 105$ GeV (10 GeV electrons, 275 GeV protons).

Table 2: Hybrid EIC-LHeC Parameters, Initial Configuration (Luminosity $\sim 2 \times 10^{33}$ cm^{-2}s^{-1})

Parameter	e	p	**Unit**
E	10.0	275.0	GeV
N	$1.19{\times}10^{10}$	$2.1{\times}10^{11}$	
$n_{bunches}$	386		
f_{collis}	3.0×10^7		Hz
Norm. ε h/v	22.0 / 22.0	3.3 / 0.3	µm
$\beta^*_{x/y}$ at IP	1.20 / 0.05	0.12 / 0.06	m
σ_z	3.0	70.0	mm
$\sigma_{x/y}$ at IP	36.7 / 7.5	36.8 / 7.8	µm
$\sigma'_{x/y}$ at IP	30.6 / 149.9	306.3 / 130.6	µrad
$\xi_{x/y}$	3.474 / 0.679	0.001 / 0.002	
I_{beam}	57.6	1000	mA
$\sigma^{eff}_{x/y}$ at IP	52.0 / 10.8		µm
Peak $\mathcal{L}$	2.1×10^{33}		cm^{-2}s^{-1}

The initial hybrid satisfies the CD0 lower boundary with unmodified EIC proton emittances; $\xi_p < 0.01$ in both configurations. The large electron ξ_e reflects severe single-pass disruption – acceptable since electrons are discarded after collision, but raising questions about ERL acceptance for the disrupted bunch (addressed in the *Energy recovery evaluations* section). The optimized case reaches 10^{34} cm^{-2}s^{-1} through emittance reductions; proton divergences up to 182 µrad

[1] Sited to avoid the Peconic River protected watershed boundary [10].

Table 3: Hybrid EIC-LHeC Parameters, Optimized Configuration (1×10^{34} cm^{-2}s^{-1}, with e-cooling and flat-beam optics). Unchanged parameters as in Table 2.

Parameter	e	p	Unit
Norm. ε h/v	22.0 / 2.2	0.9 / 0.1	µm
$\beta^*_{x/y}$ at IP	0.25 / 0.10	0.09 / 0.05	m
$\sigma_{x/y}$ at IP	16.8 / 3.4	16.8 / 3.6	µm
$\sigma'_{x/y}$ at IP	66.8 / 32.7	182.4 / 77.8	µrad
$\xi_{x/y}$	3.474 / 6.789	0.003 / 0.007	
$\sigma^{\text{eff}}_{x/y}$ at IP	23.8 / 5.0		µm
Peak $\mathcal{L}$	1.0×10^{34}		cm^{-2}s^{-1}

horizontally imply tight FD aperture requirements (see *IR optics evaluations*).

BEAM–BEAM EVALUATIONS

Beam–beam collisions, highly asymmetrical for EIC hybrid, can be evaluated using existing codes, but can be also studied using vibe-coding generated code. A simplified 2D beam–beam code was written by Claude AI in a few minutes from a paragraph-long prompt – see the Python code at [8] and beam–beam simulation illustration in Fig. 2.

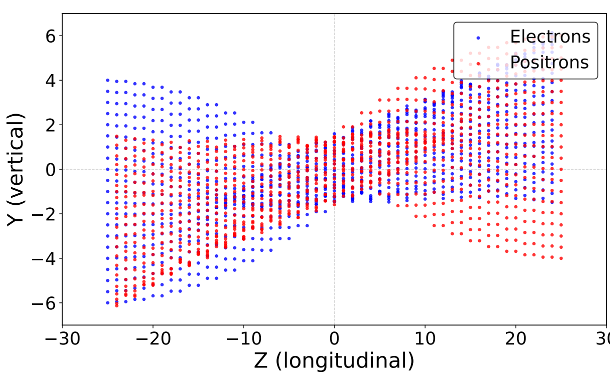

Figure 2: Example of beam–beam collision calculated by the code produced via vibe-coding. Arbitrary units in axes.

The beam–beam simulation code was generated from a single paragraph-long prompt, reproduced here to illustrate the level of physical detail required to obtain useful AI-generated code. The prompt specified: a 2D head-on collision (longitudinal Z and vertical Y coordinates only); each bunch represented by 50 transverse slices of 21 particles, placed equidistantly along Z with spacing $dZ = 1$ and vertical particle separation $dY = 0.5$; zero initial transverse velocities; and a simplified model for the beam–beam force in which the vertical electric field acting on a particle at position Y_1 equals the number of particles of the opposing slice with $Y > Y_1$ minus those with $Y < Y_1$ – the thin-slab approximation appropriate when the horizontal extent of the bunch is much larger than its vertical size. The kick coefficient D_Y was requested to be tuned so that a particle undergoes approximately one betatron oscillation during the full collision. Visualization of the collision dynamics was requested from the outset. The AI (Claude) produced working, animated code on the first attempt. Five short follow-up prompts then corrected the kick sign, tuned D_Y, added offset collisions, and improved the visualization display. The entire exchange – initial prompt plus five follow-up adjustments – took a few minutes and required no debugging of code syntax by the student. The prompt and the resulting Python code are available at [8].

This capability suggests a transformative shift in how students learn. Rather than passively attending lectures or running decades-old legacy codes, students could actively create simplified simulation codes, then iteratively refine them through AI-assisted modifications to explore different physics phenomena.

For the specific example of EIC-LHeC hybrid mini-project, students could progressively enhance their understanding by first converting the code to use of physical units; then implementing realistic beam distributions with emittance and beta-functions; adjusting input parameters to match those of EIC-LHeC hybrid; adding luminosity calculations; investigating how luminosity depends on asymmetry of disruption parameters and beam offsets; measuring angular distribution of highly disrupted electron beam.

IR OPTICS AND MAGNETS

Approach and AI prompting strategy. The IR optics design focused on the Final Doublet (FD), the pair of quadrupoles that converts the diverging beam emerging from the IP into a parallel beam downstream. The prompt specified: use thick-lens transfer matrices; trace four rays with zero position at the IP and initial angles equal to $\pm\sigma'_{x,y}$ from Table 2; assume $L^* = 2$ m (IP to entrance of Q1), quadrupole length $L_q = 2$ m, and inter-quadrupole gap 0.5 m; solve for the quadrupole strengths such that the beam exits the FD as a parallel beam in both planes simultaneously (condition $M_{22} = 0$ in the full transfer matrix); try both QF–QD and QD–QF polarities and select the one that minimises the maximum beam size inside the FD; plot X and Y beam envelopes and save the figures to PNG files. Claude generated working Python code in a single pass, requiring no manual debugging [8], and independently reproduced by Gemini.

FD design results. For the beam with IP parameters from Table 2, for the proton beam the solver selects QF–QD polarity with $k_1 = +0.319$ m^{-2} and $k_2 = -0.167$ m^{-2}, corresponding to physical gradients $G_1 = 293$ T/m and $G_2 = 153$ T/m (for $B\rho = 917$ T·m). For the electron beam the complementary QD–QF polarity is optimal, giving $G_1 = 10.6$ T/m and $G_2 = 5.6$ T/m (at $B\rho = 33.4$ T·m). The maximum beam envelope inside the FD (shown in Fig. 3) reaches 0.82 mm and 1.43 mm (horizontal and vertical) for protons; and 0.34 mm and 0.40 mm for electrons.

Magnet technology assessment. For a quadrupole of gradient G, the maximum achievable aperture radius for a given superconducting (SC) technology with peak bore field B_{bore} is $R_{\max} = B_{\text{bore}}/G$. The aperture-to-beam-size ratios $N_X = R_{\max}/\sigma_x^{\max}$ and $N_Y = R_{\max}/\sigma_y^{\max}$ quantify

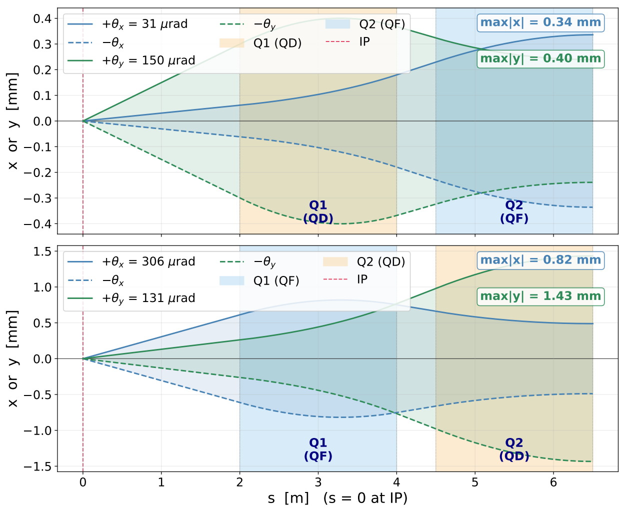

Figure 3: Hybrid EIC-LHeC FD produced via vibe-coding. Electrons (10 GeV, top plot) and protons (275 GeV, bottom).

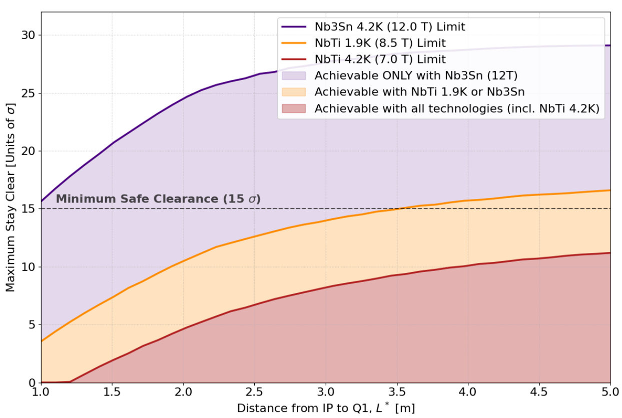

Figure 4: Hybrid EIC-LHeC FD stay-clear vs L^* and SC magnet technology analyzed via vibe-coding (with 19 mm shielding and internals subtracted).

the available margin; in collider IR practice $N_{X,Y} \gtrsim 15$ is required to accommodate orbit offsets, beam halo, and tolerances. Assuming that magnet technologies, NbTi at 4.2 K, NbTi at 1.9 K, and Nb_3Sn at 4.2 K, allow peak bore field of 7 T, 8.5 T and 12 T correspondingly, the EIC-LHeC hybrid proton FD (the most constrained) was analyzed with Gemini AI, varying the L^* parameter. The resulting beam stay clear is shown in Fig. 4, indicating the preference for NbTi at 1.9 K. For the electron FD, the required gradients are so low that SC magnets are in principle unnecessary; however SC approach may be required for practical purposes of integration within tightly constrained FD.

Further steps, within the students' EIC-LHeC mini-project, can include varying FD quadrupole lengths and other FD parameters, as well as evaluating the aperture-to-beam-size ratio for the electron beam disrupted after IP collision.

ENERGY RECOVERY EVALUATIONS

Approach and AI prompting strategy. A detailed prompt was given to Claude specifying all relevant beam parameters for the EIC-LHeC hybrid: 1.5 nC bunch charge, 1.5 mm bunch length, 7 MeV injection energy, 10 GeV SRF linac gain, 801.58 MHz RF frequency, and 40.08 MHz bunch repetition rate, yielding 60 mA in the arcs and 120 mA in the linac (accelerating plus decelerating bunches). The prompt requested a comprehensive simulation of a single-pass ERL and a two-pass ERL (5 GeV per pass), including: longitudinal phase-space tracking through acceleration, an electron-proton beam–beam collision, 180-degree phase reversal in the return arc, and deceleration; energy recovery efficiency as a function of deceleration-phase error; a detailed wall-plug power budget; and comparison with a conventional storage ring.

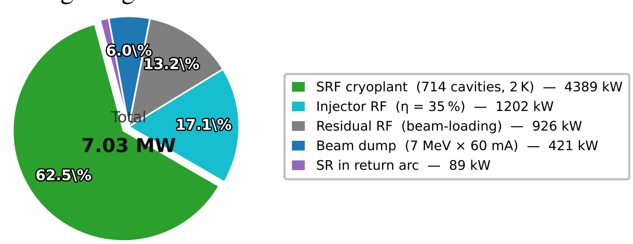

Figure 5: Single-pass ERL wall-plug power budget (10 GeV, 60 mA, f_{RF} = 801.58 MHz) produced by vibe-coding.

Energy recovery results. The AI model produced a code at the first iteration, reach with physics, including Yokoya-Chen beam–beam approximation, SR losses, comparison of single or two pass ERL with rings of different size – offering vast opportunities for discussion with students about validity and applicability of approaches. The simulation confirms an energy recovery efficiency of 99.93%, reducing the wall-plug power from 602 MW (no recovery) to 7.0 MW. Figure 5 shows the single-pass power budget breakdown.

COST ESTIMATIONS

Cost estimates for the hybrid electron complex, IR, and detector were developed using AI-assisted parametric scaling from the LHeC study [9], excluding upgrades to proton infrastructure. Two linac configurations were compared: a single-pass 10 GeV SRF linac and a recirculating 6 GeV linac traversed twice (or three times for 18 GeV), analogous to CEBAF and the LHeC design. The recirculating option halves the dominant SRF, civil, and cryo costs while adding modest arc costs (70 MCHF, 100 PY), saving ~62 MCHF and ~250 person-years (PY) overall – making it the preferred option. Estimated totals are: electron complex 278 MCHF (451 PY), IR 50 MCHF (50 PY), detector 250 MCHF (100 PY), giving a grand total of ~578 MCHF (601 PY). These figures were scaled with AI assistance and should be regarded as order-of-magnitude only; IR and detector line items were manually increased above the raw AI output.

Converting to USA DOE accounting (1 CHF = 1.25 USD; labor at 300k USD/PY; 60% contingency reflecting the preliminary nature of the estimate) yields the summary in Table 4. The total of ~1.4 B USD is approximately half the upper bound of the CD0-approved EIC cost range – a factor-of-two margin that justifies more detailed evaluation of the hybrid concept.

Table 4: USA DOE Cost Accounting, EIC-LHeC Hybrid (Recirculating 6 GeV Linac)

Item	**Value (MUSD)**
Material costs (578 MCHF × 1.25)	722.2
Labor costs (601 PY × 300k USD)	180.5
Pre-contingency total	902.7
Contingency (60%)	541.6
Total with contingency	**1444.3**

A method for more advanced cost optimization for this hybrid machine that could be investigated by students is that used by the plasma–RF hybrid collider concept HALHF [12]. This method, described in Refs. [13, 14], first builds a simplified physics model of all the major subsystems of the collider, then uses a small number of key input parameters (e.g., energies, collision frequency, accelerating gradients, etc.) to calculate lengths, power and other requirement for each subsystem, and finally converts this to an overall cost via cost estimates from similar machines. The global cost optimum is then found (approximately) using Bayesian optimization; a computationally economical machine-learning optimization method. While for HALHF this was performed using the start-to-end simulation framework ABEL [15, 16], a more simplified code for the same purpose is available at [17] – the latter being more suitable for AI training and code generation.

REQUIRED R&D AND NEXT STEPS

The hybrid EIC-LHeC concept rests on several assumptions requiring dedicated R&D before the design can be considered credible at project-proposal level. The principal challenges, identified in dialogue with AI, are the following. *High-current ERL operation:* fractional energy loss $\lesssim 10^{-4}$ is required at 58 mA and 10 GeV; a staged program at CEBAF (high energy, low current) and PERLE (low energy, high current) would bound the two critical parameters. *Polarized source current:* 58 mA far exceeds the state of the art (~1 mA CW); multiplexing N sources via RF mergers is a plausible path once 10 mA is demonstrated at Jefferson Lab. *IR final doublet:* small proton β^* (0.09–0.12 m) drives IP divergences of ~300 μrad, requiring large-aperture SC quadrupoles, tight collimation, and MDI compatibility with ePIC. *Intra-beam scattering (IBS):* the factor-of-three increase in proton bunch charge raises the IBS rate; electron cooling must maintain the emittances assumed in the 10^{34} configuration. *Disrupted-beam energy recovery:* with $\xi_y^e \approx 6.8$, the electron bunch is severely disrupted; whether the resulting phase space fits within ERL return-arc acceptance requires dedicated simulation.

Logical next steps are: beam–beam parameter scans with strong-strong codes; conceptual IR lattice and MDI design; ERL lattice and disrupted-beam acceptance study; IBS and cooling-power modeling; bottom-up cost re-estimation; and programmatic compatibility assessment with the EIC CD schedule. The hybrid concept illustrates how AI-assisted exploration can rapidly map an alternative design space and identify the critical questions – accelerating the early conceptual phase without replacing rigorous engineering analysis.

VERIFICATION OF AI RESULTS

Discuss how verification of AI results will be performed, and how it is connected with our believe that students will learn accelerator physics from such educational approach.

DRAFT FOR EVALUATION SECTION:

For instance, referring to the above mentioned beam-beam collision example, students could progressively enhance their understanding by making simulation more and more realistic, but most importantly...

Critically, students would develop diagnostic tests to verify that physics principles are correctly implemented – learning to validate AI-generated code against their understanding of the underlying physics.

This approach ensures that as we leverage AI tools, we continue developing human experts capable of designing and building the accelerators and colliders of tomorrow. The goal isn't to replace human expertise, but to accelerate its development while maintaining rigor and deep understanding.

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CONNECTION TO GENESIS PROGRAM

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SELF-ASSESSMENT AND NEXT STEPS

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This section replaces Conclusion section

(NOTE ON PAPER LENGTH: THIS IS INVITED POSTER, SO THE LIMIT IS 5 PAGES. REFERENCES CAN BE ON EXTRA PAGE.)

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